Leaf Vein Quantification and Measurements

叶脉定量与测量

Sentence by sentence bilingual version | 逐句中英文对照版

No.	English original	中文翻译
1	**EN:** For leaf vein imaging, expanded flag leaves were collected from the glasshouse before and after heat treatments and fixed in Formalin Acid Alcohol (FAA) solution (composed of 37% aqueous formaldehyde solution, 70% ethanol, and 5% glacial acetic acid, pH 7.0) for later use.	**中文：** 为进行叶脉成像，在热处理前后从温室中采集完全展开的旗叶，并将其固定于福尔马林 乙酸 酒精固定液（FAA）中备用。该固定液由 37% 甲醛水溶液、70% 乙醇和 5% 冰乙酸组成，pH 为 7.0。
2	**EN:** Leaf samples were prepared for imaging following the clearing and staining protocol (Naznin et al., 2026; Scoffoni & Sack, 2013).	**中文：** 叶片样品按照透明化和染色流程进行成像前处理，方法参考 Naznin et al. (2026) 以及 Scoffoni and Sack (2013)。
3	**EN:** The widest part of each leaf was excised in 1 cm lengths for free-hand sectioning.	**中文：** 每片叶取最宽部位，切取 1 cm 长的叶段，用于徒手切片。
4	**EN:** Leaf sections were cleared using a 10% NaOH aqueous solution, soaking the leaf sections in it for two days.	**中文：** 叶段使用 10% NaOH 水溶液进行透明化处理，并在该溶液中浸泡两天。
5	**EN:** Once the leaves appeared transparent and softened, the NaOH solution was removed, and the leaf sections were rinsed three times with water.	**中文：** 待叶片呈透明并软化后，去除 NaOH 溶液，并用清水冲洗叶段三次。
6	**EN:** Subsequently, the leaves were dehydrated using an ethanol (EtOH) dilution series (30%, 50%, 70%, 100%), with each change taking no more than 5 minutes, followed by removing of the liquid.	**中文：** 随后，使用乙醇（EtOH）梯度对叶片进行脱水处理，浓度依次为 30%、50%、70% 和 100%。每次换液时间不超过 5 分钟，之后去除液体。
7	**EN:** After the 100% EtOH stage, the leaf samples were immersed in 1% safranin (1 g safranin/100 ml of 100% EtOH) for 10 minutes.	**中文：** 完成 100% EtOH 步骤后，将叶片样品浸入 1% 番红溶液中染色 10 分钟。该溶液的配制方法为每 100 ml 100% EtOH 中加入 1 g 番红。
8	**EN:** Then, the samples were gently rinsed with 100% EtOH and vacuumed	**中文：** 随后，用 100% EtOH 轻轻冲洗样

	off the liquid.	品，并通过真空方式去除液体。
9	**EN:** Next, the samples were covered in 1% fast green (1 g fast green/100 ml of 100% EtOH) for a few seconds, followed by gentle rinsing with 100% EtOH and vacuuming off the liquid.	**中文：** 接着，将样品覆盖于 1% 固绿溶液中数秒。该溶液的配制方法为每 100 ml 100% EtOH 中加入 1 g 固绿。随后，用 100% EtOH 轻轻冲洗，并通过真空方式去除液体。
10	**EN:** The samples were then brought back to water through a reverse dilution series.	**中文：** 之后，样品通过反向稀释梯度重新转入水相。
11	**EN:** After the water stage, the leaves were mounted on slides using water for vein imaging.	**中文：** 进入水相阶段后，以水作为封片介质将叶片置于载玻片上，用于叶脉成像。
12	**EN:** The leaves were imaged under x10 magnification using a CCD camera (NIS-F1 Nikon, Tokyo, Japan) attached to a light microscope (Leica Microsystems AG, Germany).	**中文：** 叶片在 10 倍放大倍数下进行成像，使用连接于光学显微镜（Leica Microsystems AG，德国）的 CCD 相机（NIS-F1 Nikon，日本东京）拍摄。
13	**EN:** The images were analysed using ImageJ software (Al-Salman et al., 2023).	**中文：** 图像使用 ImageJ 软件进行分析，参考 Al-Salman et al. (2023)。
14	**EN:** The anatomical characteristics evaluated included vein density (VD), thickness of major vein and interveinal distance (IVD).	**中文：** 测定的解剖学特征包括叶脉密度（VD）、主脉厚度以及脉间距（IVD）。
15	**EN:** VD was calculated as the total length of longitudinal veins per unit leaf area (mm mm^{-2}), and IVD (μm) was defined as the average distance between consecutive minor longitudinal veins.	**中文：** VD 按单位叶面积内纵向叶脉总长度计算，单位为 mm mm^{-2}。IVD 的单位为微米（μm），定义为相邻小纵向叶脉之间的平均距离。

Key terminology | 术语对照

English	中文
expanded flag leaves	完全展开的旗叶
Formalin Acid Alcohol (FAA)	福尔马林 乙酸 酒精固定液（FAA）
clearing and staining protocol	透明化和染色流程
free-hand sectioning	徒手切片
ethanol dilution series	乙醇梯度
safranin	番红
fast green	固绿

vein density (VD)	叶脉密度（VD）
thickness of major vein	主脉厚度
interveinal distance (IVD)	脉间距（IVD）

Note: This version restructures the original methods paragraph into sentence aligned English Chinese pairs for easier reading and checking.

Supplementary notes added to the sentence by sentence version
逐句中英文对照版新增注意点

Purpose: the earlier short PDF mainly translated the wheat leaf vein quantification method sentence by sentence. The uploaded original bilingual protocol is broader. It contains background, terminology, measurement formulas, practical checklists, barley specific notes, solution preparation order, and bench side workflow.
目的：前一版短 PDF 主要是对小麦叶脉量化方法进行逐句中英文对照翻译。你上传的原先协议范围更大，包含背景、术语、计算公式、实验前清单、针对大麦的实操提醒、配液顺序和实验当天操作单。

1. Main additions and substantive differences
主要新增内容与实质性差异

Aspect 比较点	Short wheat method PDF 短版小麦方法	Original broader protocol 原先完整协议	What should be added or noted 建议补充或注意
Document scope 文件范围	Focuses on fixation, clearing, staining, imaging, and ImageJ analysis for wheat flag leaves. 重点是小麦旗叶的固定、透明化、染色、成像和 ImageJ 分析。	Adds background, definitions, measurement formulas, preparation checklists, barley practical notes, solution preparation order, and bench side workflow. 增加了背景、定义、计算公式、实验准备清单、大麦实操建议、配液顺序和实验当天操作单。	Keep the short method as the manuscript style method, but add a practical appendix for laboratory execution. 短版方法可作为论文方法部分，另加一个实操附录更合适。
Species logic 物种逻辑	Uses wheat flag leaves before and after heat treatment. Wheat is a monocot grass with mainly parallel venation. 使用热处理前后的小麦旗叶。小麦是单子叶禾本科植物，主要为平行脉。	The original explanatory framework includes dicot leaf vein order concepts, and the appendix reminds that barley is also a monocot grass with parallel venation. 原协议正文包含双子叶叶脉等级概念，附录提醒大麦也是平行脉为主的禾本科植物。	For wheat, do not force all dicot style metrics. Focus on longitudinal vein density, interveinal distance, and major vein thickness. 小麦不应强行套用所有双子叶指标，应重点关注纵向叶脉密度、脉间距和主脉厚度。
Measurement scope 测量范围	Reports vein density, major vein thickness, and interveinal distance. 报告叶脉密度、主脉厚度和脉间距。	Also discusses leaf area, perimeter, length to width ratio, vein order density, minor vein density, total vein density, and free vein endings. 还讨论叶面积、周长、长宽比、各级叶脉密度、细脉密度、总叶脉密度和自由末端叶脉。	Optional metrics can be mentioned only if they are actually measured. The core wheat method should remain VD, IVD, and major vein thickness. 可选指标只有在实际测量时才加入。小麦核心方法仍建议保留 VD、IVD 和主脉厚度。
Practical execution 实操层面	The short method gives the main procedure but few trial or troubleshooting instructions. 短版方法给出了主要流程，但预实验和排错说明较少。	Adds checklists, chemical lists, first trial strategy, record keeping, and safety reminders. 增加了清单、化学品列表、第一次预实验策略、记录建议和安全提醒。	Add a concise checklist, especially for NaOH clearing, staining time, imaging position, and ImageJ scale calibration. 建议补充简短清单，尤其是 NaOH 透明化、染色时间、成像位置和 ImageJ 比例尺校准。

Key conclusion: the original broader protocol is not only a longer translation. Its most useful addition is practical control of experimental variability. The short wheat PDF needs three types of additions: parameter comparison, species specific measurement logic, and execution checklist.
核心判断：原先完整协议不只是更长的翻译。它最有用的地方是帮助控制实验误差。短版小麦 PDF 最需要补充三类内容：参数差异说明、禾本科叶片适用指标判断，以及实验执行清单。

Supplementary notes added to the sentence by sentence version
逐句中英文对照版新增注意点

2. Method parameters that should not be mixed directly
不能直接混用的方法参数

Step 步骤	Current wheat method 当前小麦方法	Broader protocol 完整协议中对应内容	Practical note 实际注意点
Fixation 固定	FAA solution: 37% aqueous formaldehyde solution, 70% ethanol, and 5% glacial acetic acid, pH 7.0. FAA 固定液：37% 甲醛水溶液、70% 乙醇和 5% 冰醋酸，pH 7.0。	Formalin acetic acid storage solution: 37% aqueous formaldehyde solution, 50% ethanol, and 13% glacial acetic acid solution. 福尔马林醋酸保存液：37% 甲醛水溶液、50% 乙醇和 13% 冰醋酸溶液。	Corrected note: FAA means Formalin Acid Alcohol. It is the same type of fixative concept, not a different method in principle. The apparent difference is mainly how the recipe is written. For the manuscript, keep the exact FAA recipe that your lab actually used. Use the broader protocol only as a practical reference for optimisation. 更正说明：FAA 指 Formalin Acid Alcohol，即福尔马林 乙酸 酒精固定液。从概念上看，它不是另一套完全不同的方法。表面差异主要来自配方书写方式。论文中应保留你们实验室真实使用的 FAA 配方；完整 protocol 主要作为后续预实验优化参考。
Clearing 透明化	10% NaOH aqueous solution for two days, until sections become transparent and softened. 10% NaOH 水溶液处理两天，直到叶片透明并软化。	5% NaOH for several hours to several days, or 2.5% NaOH for delicate tissues. 5% NaOH 处理数小时到数天，脆弱组织可用 2.5% NaOH。	This is a major difference. The wheat method is stronger. In future trials, monitor tissue softness and image quality rather than relying only on fixed time. 这是很大的差异。小麦方法更强。后续预实验要观察组织软化程度和图像质量，不能只依赖固定时间。
Bleaching 漂白	No bleach step is included in the current wheat method. 当前小麦方法没有漂白步骤。	50% bleach can be used for 10 to 20 seconds if colour remains. 如果仍有颜色残留，可使用 50% 漂白液处理 10 到 20 秒。	Do not add bleach routinely unless staining or clearing fails. Bleach may over decolourise or damage grass leaf tissues. 不要常规加入漂白步骤，除非透明化或染色效果不理想。漂白可能导致过度脱色或损伤禾本科叶片组织。
Staining 染色	1% safranin for 10 minutes, then 1% fast green for a few seconds. 1% 番红染色 10 分钟，随后 1% 固绿染色数秒。	1% safranin for 2 to 30 minutes depending on species, then 1% fast green for a few seconds. 1% 番红根据物种染色 2 到 30 分钟，随后 1% 固绿染色数秒。	The broader protocol supports optimisation. For wheat, keep 10 minutes if it worked, but record actual staining response and adjust only after pilot testing. 完整协议说明染色时间可优化。对于小麦，如果 10 分钟效果好就保留，但要记录染色反应，并只在预实验后调整。
Imaging 成像	Images are taken under 10x magnification using a CCD camera attached to a light microscope. 使用连接光学显微镜的 CCD 相机，在 10 倍放大下成像。	The broader protocol suggests scanning whole leaves and using 5x or 10x microscopy for different vein traits. 完整协议建议整叶扫描，并根据不同叶脉性状使用 5 倍或 10 倍显微成像。	For the wheat paper, report the actual equipment and 10x magnification. Whole leaf scanning is optional only if leaf level traits are added. 小麦论文应报告实际设备和 10 倍放大。只有增加整叶水平指标时，才需要考虑整叶扫描。

3. The most important species specific adjustment
最重要的物种适配

The broader protocol contains many vein order concepts that are mainly useful for dicot reticulate venation. Wheat and barley are monocot grasses, so their leaves mainly show parallel venation. Therefore, the practical focus should be longitudinal vein density, interveinal distance, vascular bundle or major vein thickness, and consistent imaging positions.
完整协议中有很多叶脉等级概念，这些内容主要适用于双子叶植物的网状叶脉。小麦和大麦都属于单子叶禾本科植物，叶片主要是平行脉。因此，真正适合小麦的重点应是纵向叶脉密度、脉间距、维管束或主脉厚度，以及固定一致的成像位置。

Free ending veins, fourth to seventh order veins, and dicot style vein hierarchy should not be forced into the wheat method unless the image structure clearly supports those measurements and the study objective requires them.
自由末端叶脉、四级到七级叶脉、以及双子叶式叶脉等级体系，不建议强行加入小麦方法。只有当图像结构确实支持这些测量，并且研究目标需要时，才考虑增加。

Supplementary notes added to the sentence by sentence version
逐句中英文对照版新增注意点

4. Practical checklist to add at the end
可直接加在文末的实操清单

Item 项目	Recommended note to add 建议加入的注意点
Sample consistency 样品一致性	Use fully expanded flag leaves collected from the same developmental stage and comparable positions. Record genotype, treatment, replicate, sampling date, and whether the leaf was collected before or after heat treatment. 使用处于相同发育阶段和相近位置的充分展开旗叶。记录基因型、处理、重复、取样日期，以及样品来自热处理前还是热处理后。
Pilot test 预实验	Before processing all samples, run a small pilot test to check whether the 10% NaOH two day clearing step causes over softening, tearing, or poor staining. If tissue damage occurs, optimise clearing time or NaOH strength before scaling up. 正式处理所有样品前，先做小规模预实验，确认 10% NaOH 两天透明化是否会导致叶片过软、破损或染色效果差。如果组织受损，应先优化透明化时间或 NaOH 浓度，再扩大样品量。
Imaging position 成像位置	Keep imaging positions consistent across all leaves. For wheat flag leaves, the widest part or a predefined central region should be used consistently. Avoid damaged edges, folded tissue, and areas with uneven staining. 所有叶片的成像位置必须保持一致。对于小麦旗叶，建议固定使用最宽部位或预先定义的中央区域。避免边缘破损、组织折叠和染色不均的区域。
ImageJ calibration ImageJ 校准	Set the image scale before measurement. Save the original image, the calibrated image, and the annotated measurement file. Report the unit clearly, for example VD in mm mm^-2 and IVD in micrometre. 测量前必须在 ImageJ 中设置比例尺。保存原始图像、校准后的图像和带标注的测量文件。单位要明确，例如 VD 使用 mm mm^-2，IVD 使用微米。
Measurement rule 测量规则	Define how longitudinal vein length, major vein thickness, and interveinal distance are measured before analysis. Use the same rule for all images and record any excluded images with reasons. 分析前先明确纵向叶脉长度、主脉厚度和脉间距的测量规则。所有图像使用同一规则，并记录被排除图像及其原因。
Safety 安全	FAA, formaldehyde, glacial acetic acid, NaOH, ethanol, safranin, and fast green require appropriate PPE, waste handling, and laboratory approval. Check local SDS before starting. FAA、甲醛、冰醋酸、NaOH、乙醇、番红和固绿都需要合适的个人防护、废液处理和实验室审批。开始前应查阅本地 SDS。

5. Suggested wording for a final short note

可放入方法附注的英文表达

Because wheat leaves are monocot grass leaves with mainly parallel venation, vein order based traits designed for dicot reticulate venation were not forced into the analysis. The measurements were therefore focused on longitudinal vein density, major vein thickness, and interveinal distance, with consistent sampling and imaging positions across treatments.

由于小麦叶片属于单子叶禾本科叶片，主要表现为平行脉，因此本研究没有强行套用主要面向双子叶网状叶脉的叶脉等级性状。测量重点放在纵向叶脉密度、主脉厚度和脉间距，并在不同处理之间保持一致的取样和成像位置。

Supplementary note: Which version should be followed for barley?

补充说明：我是大麦实验，到底应该相信哪一个版本?

Short answer: for barley, use your current wheat or barley laboratory method as the main protocol, and use the broader leaf vein protocol only to guide optimisation and quality control. Do not blindly copy the dicot vein order measurements into barley.

直接结论：对于大麦，应以你现在这套小麦或大麦实验室方法作为主 protocol，完整 leaf vein protocol 只作为优化和质控参考。不要把双子叶植物的叶脉等级测量直接套到大麦上。

Decision point	Recommended choice for barley	Reason and practical note
FAA or formalin wording FAA 还是 formalin 的说法	Keep using FAA if your current method already says Formalin Acid Alcohol. 如果你们当前方法已经写 FAA，就继续按 FAA 写。	FAA is not simply formalin alone. It contains formaldehyde or formalin, alcohol, and acetic acid. So the key is not the name, but the actual recipe and what was used in the lab. FAA 不是单纯的福尔马林，而是含甲醛或福尔马林、酒精和乙酸的固定液。因此重点不是名字，而是实际配方和实验室真实操作。
NaOH clearing NaOH 透明化	Start from the gentler condition for barley, such as 2.5 percent NaOH in a pilot test. Move toward 5 percent only if clearing is insufficient. 大麦建议预实验先从温和条件开始，例如 2.5 percent NaOH；如果透明化不够，再考虑 5 percent。	Barley leaves are narrow grass leaves and can become too soft after clearing. Over clearing is harder to rescue than under clearing. 大麦叶片狭长，透明化后容易过软。透明化过度比透明化不足更难补救。
Bleach step 漂白步骤	Use bleach only when colour remains after NaOH clearing and rinsing. Treat it as optional, not automatic. 只有在 NaOH 透明化和冲洗后仍有明显颜色残留时才使用漂白。它是可选步骤，不是必须步骤。	Bleach can improve clarity, but it can also damage softened tissue. For barley, short exposure and careful monitoring are important. 漂白可以改善透明度，但也可能损伤已软化组织。大麦应短时间处理并密切观察。
Traits to measure 优先测哪些指标	Prioritise longitudinal vein density, interveinal distance, major vein or vascular bundle thickness, and consistent imaging positions. 优先关注纵向叶脉密度、脉间距、主脉或维管束厚度，以及一致的成像位置。	Barley is a monocot grass with mainly parallel venation. Dicot style free ending veins and fourth to seventh order vein hierarchy are usually not the main target. 大麦是单子叶禾本科植物，以平行脉为主。双子叶式自由末端叶脉和四级到七级叶脉等级通常不是主要指标。
Imaging design 成像设计	Use fixed sampling positions, for example upper, middle, and lower blade regions, or a clearly defined widest leaf region. 使用固定取样位置，例如叶片上部、中部和下部，或明确规定最宽处区域。	Consistency across genotype, treatment, and replicate is more important than copying every imaging position from the general dicot protocol. 不同基因型、处理和重复之间位置一致，比照搬通用双子叶 protocol 的所有成像位置更重要。

Recommended final wording for your barley methods:
For barley leaves, the general leaf vein clearing and staining protocol was used as a reference, but trait selection was adapted to monocot grass leaf anatomy. The analysis focused on longitudinal vein density, interveinal distance, and major vein or vascular bundle thickness, rather than dicot specific vein order traits. Clearing and staining times were optimised through small pilot tests to avoid over clearing, tissue damage, or overstaining.

建议写进方法或备注中的英文：
For barley leaves, the general leaf vein clearing and staining protocol was used as a reference, but trait selection was adapted to monocot grass leaf anatomy. The analysis focused on longitudinal vein density, interveinal distance, and major vein or vascular bundle thickness, rather than dicot specific vein order traits. Clearing and staining times were optimised through small pilot tests to avoid over clearing, tissue damage, or overstaining.